



MOIZ DAWOODI
O/A LEVEL PHYSICS

O' LEVEL PHYSICS

Simple Magnetism & Magnetic Fields

4.1

STUDENT NAME



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PHYSICS WITH
MOIZ DAWOODI

O'levels

4.1

Simple Magnetism & Magnetic Fields

SIR MOIZ DAWOODI

SYLLABUS OUTLINE:

4.1 Simple magnetism and magnetic fields

- 1 Describe the forces between magnetic poles and between magnets and magnetic materials, including the use of the terms north pole (N pole), south pole (S pole), attraction and repulsion, magnetised and unmagnetised
- 2 Describe induced magnetism
- 3 State the difference between magnetic and non-magnetic materials
- 4 State the differences between the properties of temporary magnets (made of soft iron) and the properties of permanent magnets (made of steel)
- 5 Describe a magnetic field as a region in which a magnetic pole experiences a force
- 6 Describe the plotting of magnetic field lines with a compass or iron filings and the use of a compass to determine the direction of the magnetic field
- 7 Draw the pattern and direction of the magnetic field lines around a bar magnet
- 8 State that the direction of the magnetic field at a point is the direction of the force on the N pole of a magnet at that point
- 9 Know that the relative strength of a magnetic field is represented by the spacing of the magnetic field lines
- 10 Describe uses of permanent magnets and electromagnets

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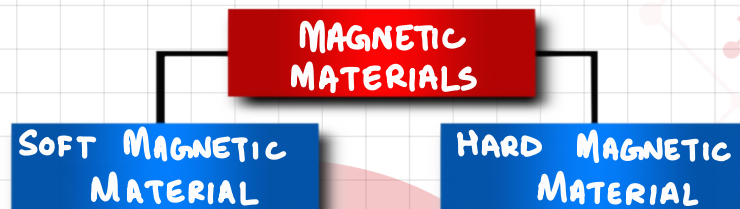


TYPES OF MATERIALS ACCORDING TO MAGNETISM:

MAGNETIC MATERIALS:

"Materials which are attracted by a magnet or can behave as a magnet are called magnetic materials."

E.g: Iron, Nickel, Cobalt, steel etc.



SOFT MAGNETIC MATERIALS:

"Magnetic materials which can be easily magnetised or easily demagnetised are called soft magnetic material."

E.g: Iron (Ferromagnet). آئرن کا کوئی دین ایمان نہیں ہوتا۔

Iron makes temporary magnet.

HARD MAGNETIC MATERIALS:

"Magnetic materials which are hard to magnetise & demagnetise are called hard magnetic material."

E.g: Steel (Sakht Londa).

Steel makes permanent magnets.

NON-MAGNETIC MATERIALS:

"Materials which can't be attracted by a magnet nor can behave as a magnet are called non-magnetic materials."

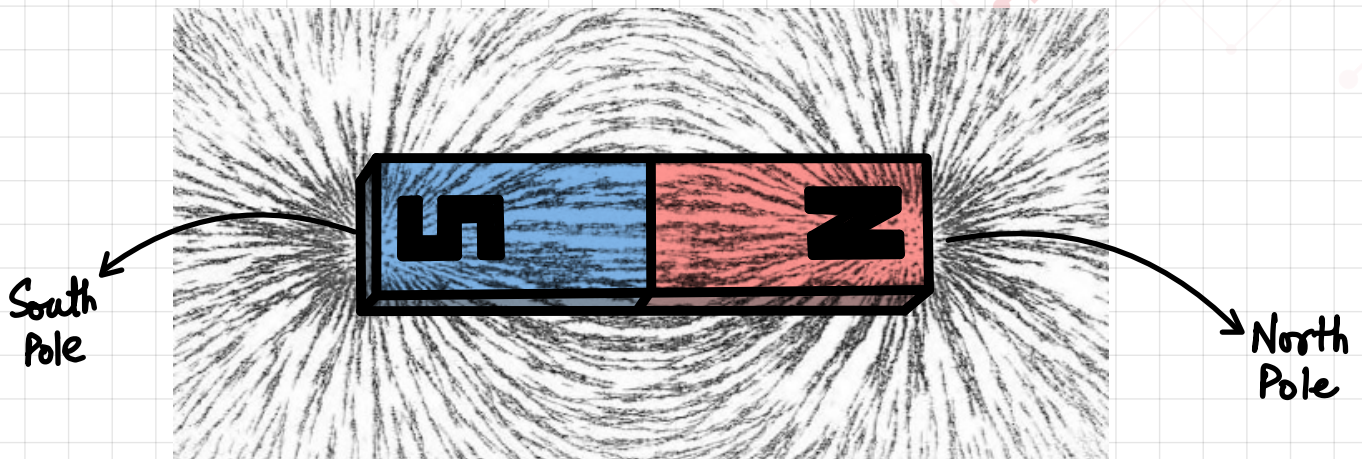
E.g: Copper, Silver, Gold, Aluminium etc.

LAWS OF MAGNETISM:

- Like poles repel.
- Unlike poles attract.

POLE:

"Part of a magnet where magnetic field is strongest is called a pole."

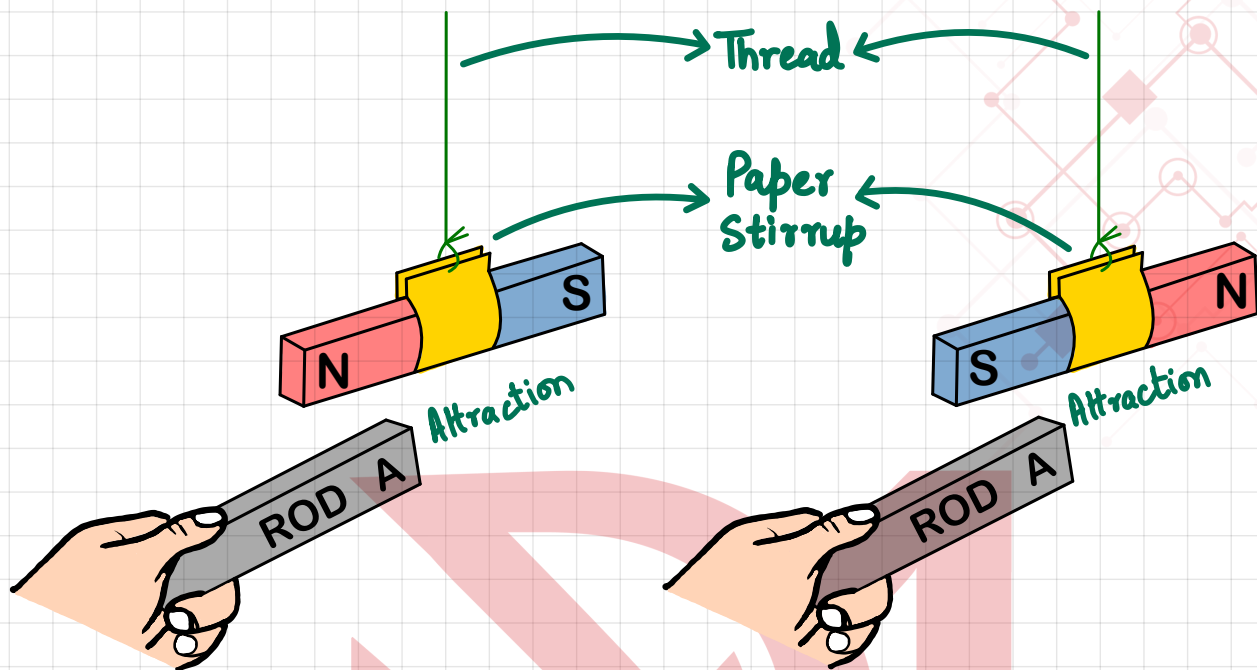


PROPERTIES OF PERMENANT & TEMPORARY MAGNETS:

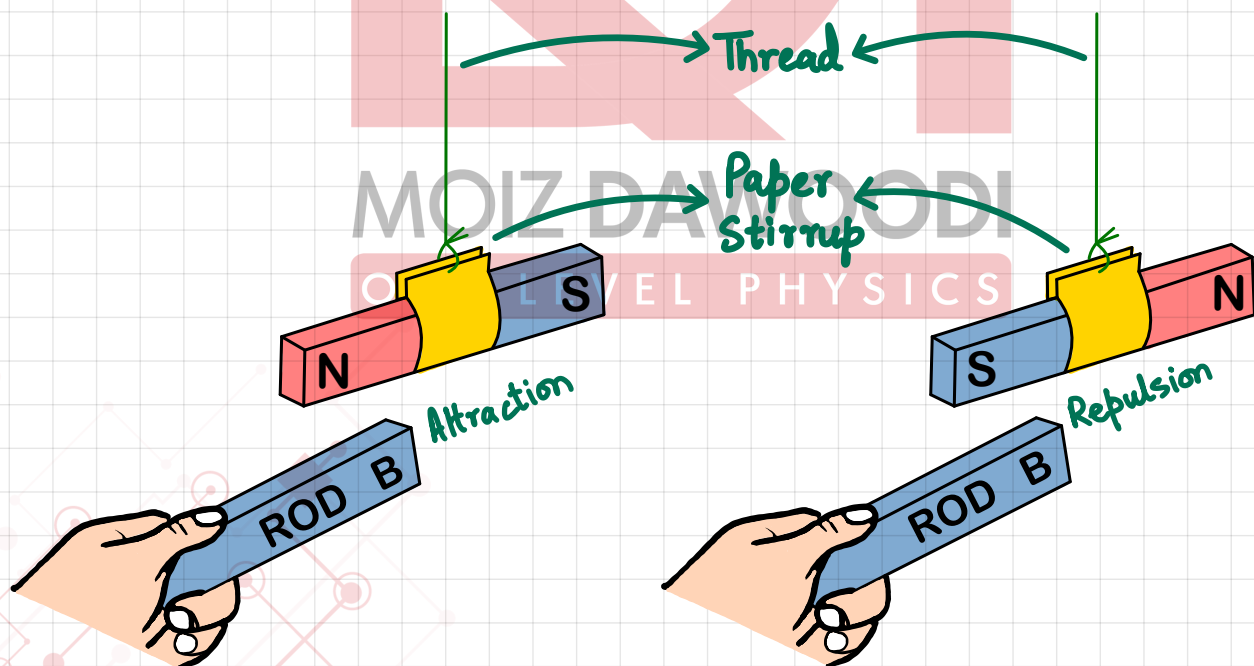
PERMENANT MAGNET	TEMPORARY MAGNET
• Magnets which retain their magnetism for longer durations. • They are hard to magnetise & hard to demagnetise. • Steel.	• Magnets which sometimes works as a magnet & sometimes as non magnet. • They are easy to magnetise & easy to demagnetise. • Iron.

How To IDENTIFY MAGNET:

BY USING PERMENANT BAR MAGNET:



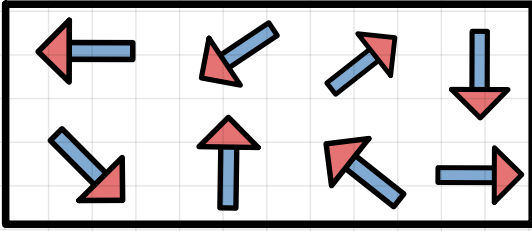
Rod A is soft magnetic material (iron) as it is attracted by both poles.



Rod B is made up of hard magnetic material (steel) as it doesn't change its poles & they shown end is South as it is repelled by south pole.

Attraction doesn't help to identify magnet. Only repulsion helps to identify magnets.

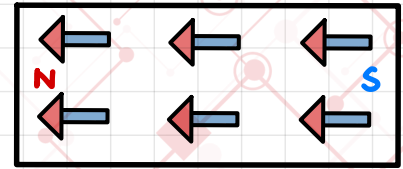
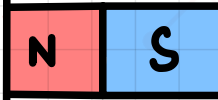
UNMAGNETISED IRON



All magnetic atomic dipoles are disaligned & cancel out each others effect.

MAGNETISED IRON

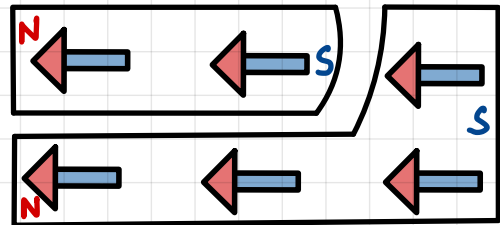
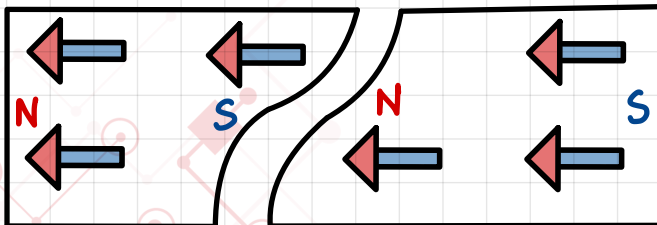
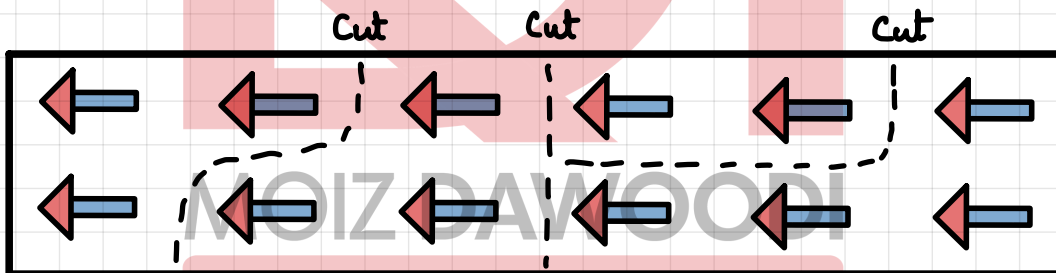
Permanent Magnet



All magnetic dipoles are aligned & support each other.

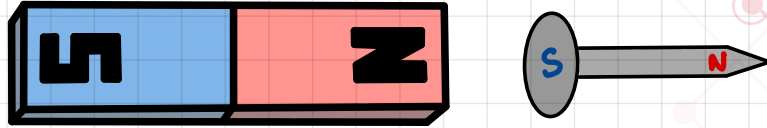
THEORY OF MAGNETISM:

If a magnet is cut down into several pieces, each part of the magnet will behave as a magnet but with lesser strength.



INDUCED MAGNETISM:

Without making physical contact, with the help of magnetic field of a bar magnet, poles are created in iron by aligning all the dipoles in specific manner.

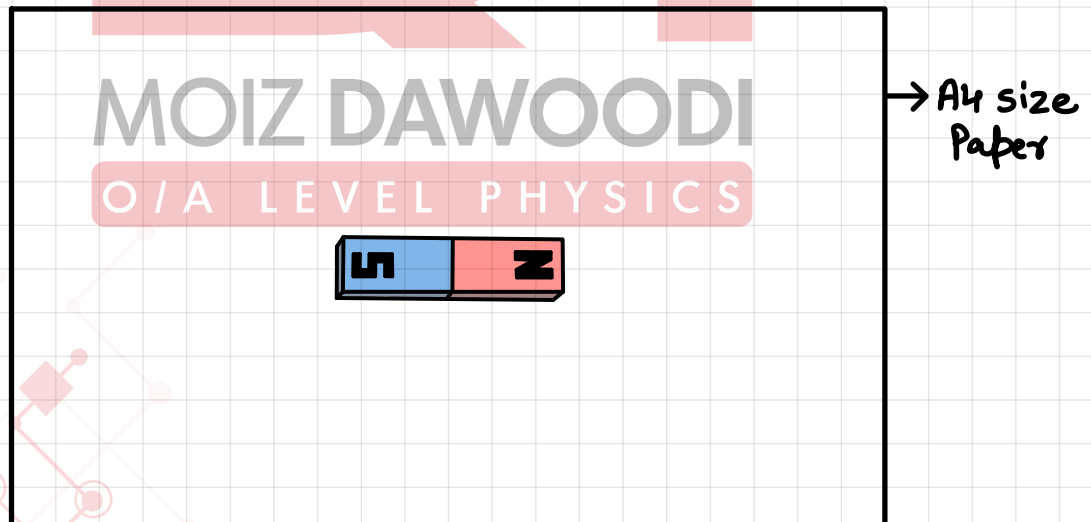


MAGNETIC FIELD:

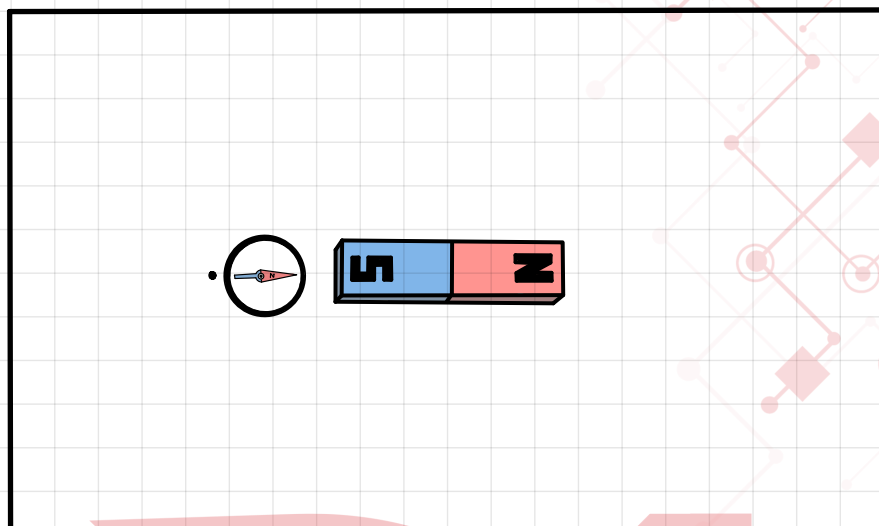
"The region around the magnet in which a magnet can experience magnetic force of other magnet is called magnetic field"

PLOTTING OF MAGNETIC FIELD LINES:

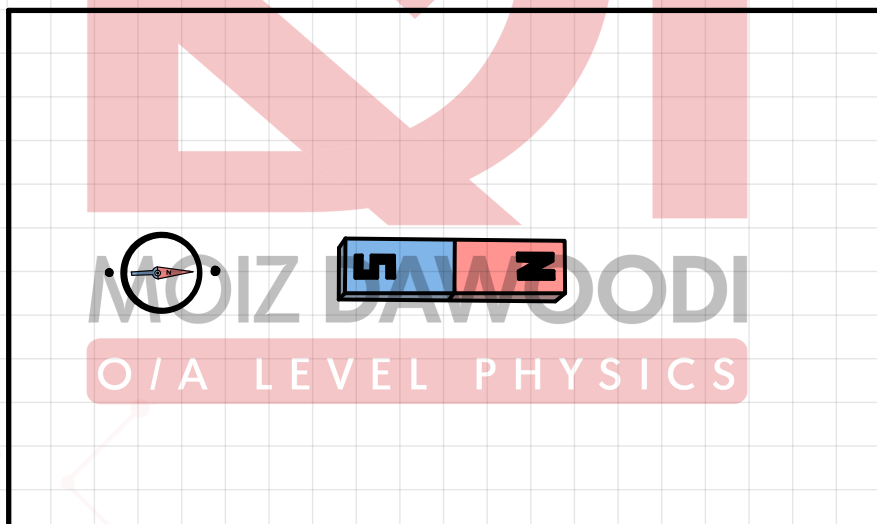
- Place bar magnet on A4 size paper & draw an outline around a bar magnet.



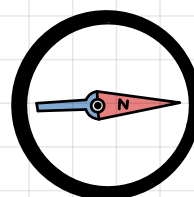
- Place magnetic compass near the pole. Give some time for magnetic compass to be aligned then mark a dot where the end of south alligns.



- Replace magnet compass such that north of compass faces dot marked earlier. Give some time for compass needle to align itself & mark second dot where south of compass alligns.



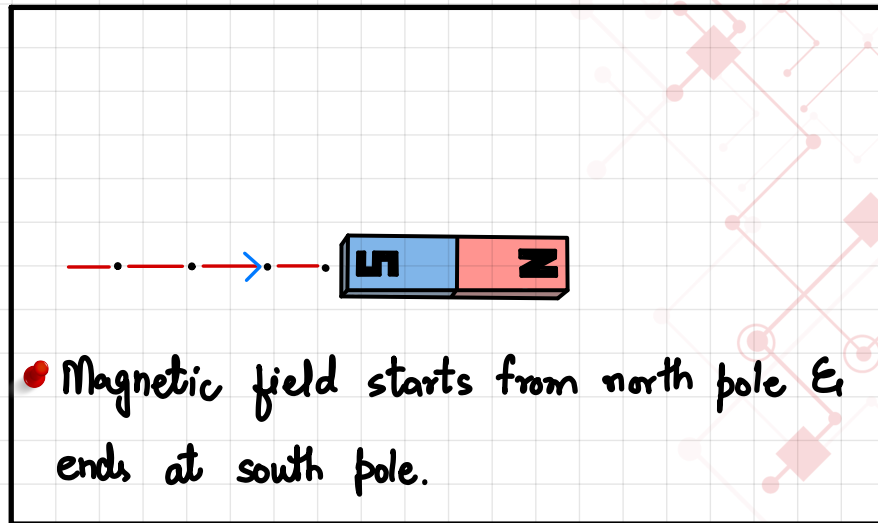
Small magnetic compass



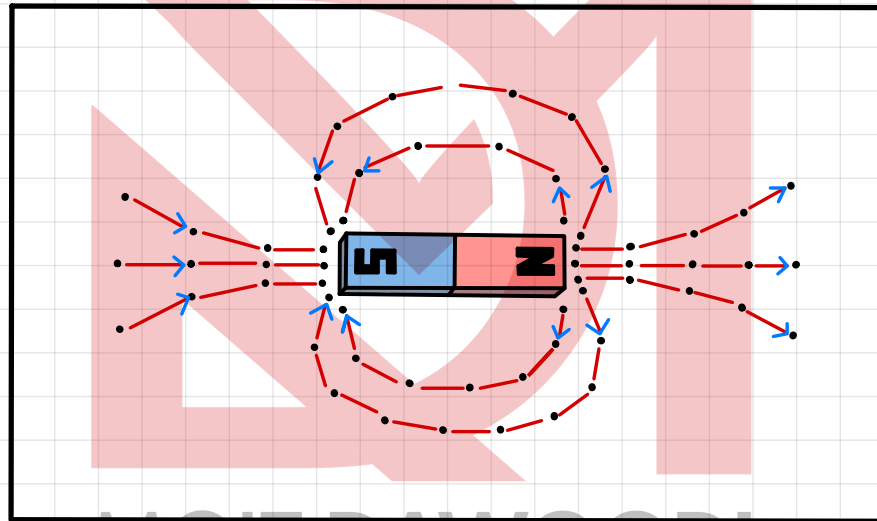
Large magnetic compass

- We always small magnetic compass as dots will be closer so accuracy of magnetic field pattern increases. Due to small magnetic needle, it can plot weakest magnetic fields as well.

- Redo this until you have enough dots. Join all the dots to draw magnetic field lines.



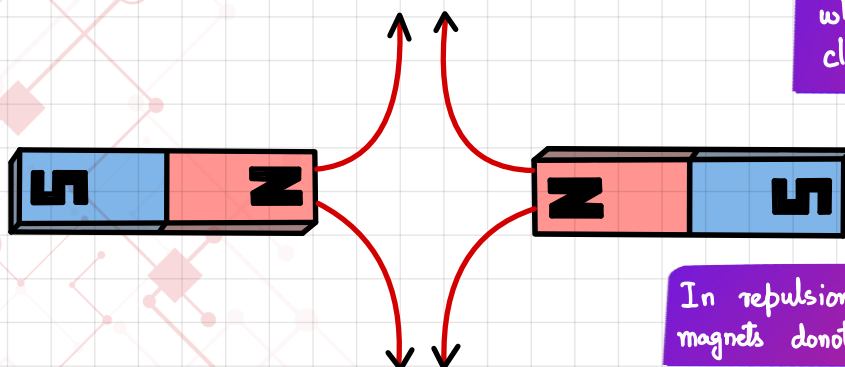
MAGNETIC FIELD PATTERN OF A BAR MAGNET:



- Direction of magnetic field lines represent the direction of magnetic force on north pole of a magnet.

PATTERN OF MAGNETIC FIELD:

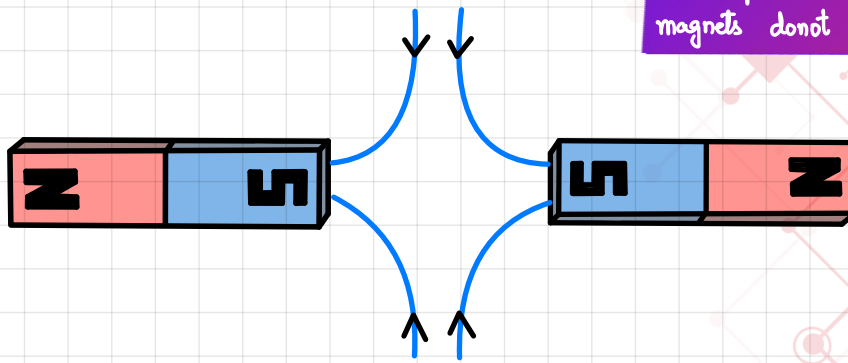
- BETWEEN TWO NORTH POLES:**



Magnetic field is stronger when field lines are closer with each other.

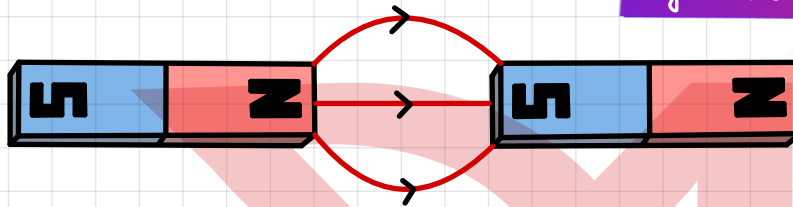
In repulsion, magnetic field lines of magnets donot meet with each other.

• BETWEEN TWO SOUTH POLES:



In repulsion, magnetic field lines of magnets do not meet with each other.

• BETWEEN NORTH & SOUTH POLE:



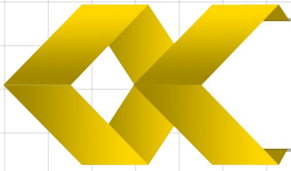
In attraction, magnetic field lines of magnets meet with each other.

USES OF PERMANENT MAGNET:

- They are used in electric motors.
- They are used in electric generators.
- They are used in magnetic compass.
- They are used in hard drives.

USES OF ELECTROMAGNETS. PHYSICS

- They are used in loudspeakers.
- They are used in electric bells.
- They are used in relays.
- They are used to magnetise or demagnetise a magnetic material.



Match the following



SCAN ME



MATCH THE FOLLOWING
TERMINOLOGIES WITH CORRECT
INFORMATION.

Soft magnetic material	are easy to magnetize & demagnetize.
Hard magnetic material	starts from north pole.
Magnetic field lines	from steel
Small magnetic compass	field lines closer with each other.
Permanent magnet is made	are hard to magnetize & demagnetize.
Iron is used to make	is a non magnetic material.
Strong magnetic field has	repel.
Copper	can be identified by repulsion.
Like poles	produces accurate magnetic field lines.
Permanent magnet	electromagnets.

Past Papers

01 Which list contains an example of a non-magnetic material, a magnetic material and a magnetised material?

A copper, iron, a compass needle

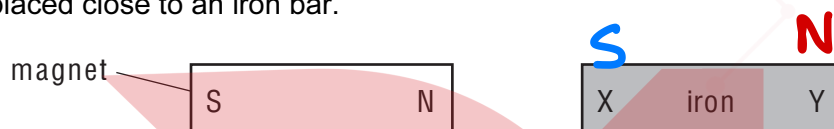
B copper, iron, polythene

C iron, steel, a compass needle

D iron, steel, polythene

[MJ2011/P11/Q26]

02 A magnet is placed close to an iron bar.



The iron bar becomes an induced magnet.

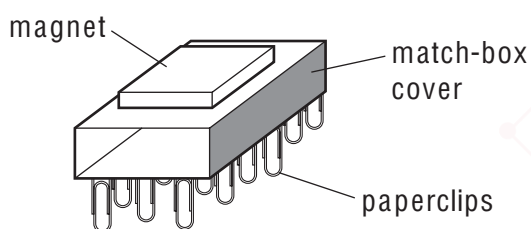
Which magnetic poles are formed at the ends X and Y of the bar?

	end X	end Y
A	N	N ✓
B	N	S
C	S ✓	N ✓
D	S ✓	S

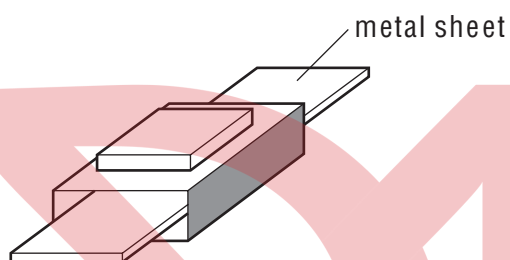
[ON2012/P12/Q26]

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- 03** A teacher sticks a magnet to the top surface of a match-box cover. The bottom surface is placed in a small tray of iron paperclips. As the match-box cover is lifted up, a large number of paperclips are held on the bottom surface.



Sheets of metal are placed inside the match-box cover, between the magnet and the paperclips.



Copper/Aluminium

iron

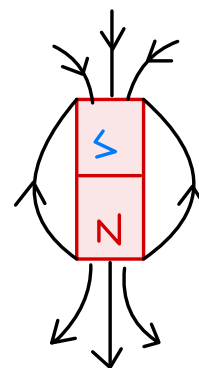
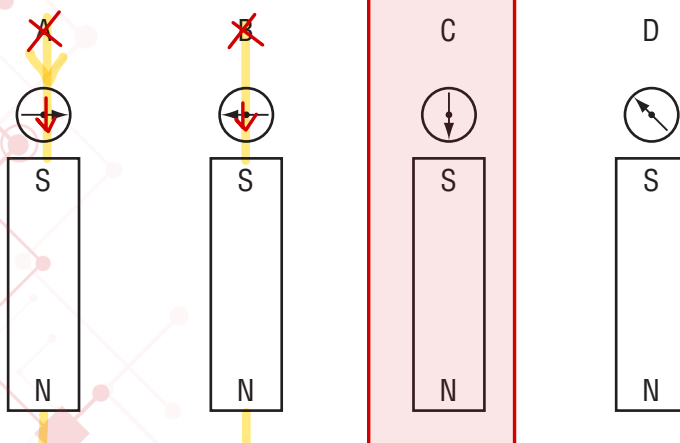
When **sheet X** is placed **inside**, the **paperclips stay attached**. When **sheet Y** is placed inside, all the **paperclips fall off**.

Which metals are the sheets made from?

	sheet X	sheet Y
A	aluminium ✓	copper
B	copper ✓	iron ✓
C	iron	aluminium
D	iron	copper

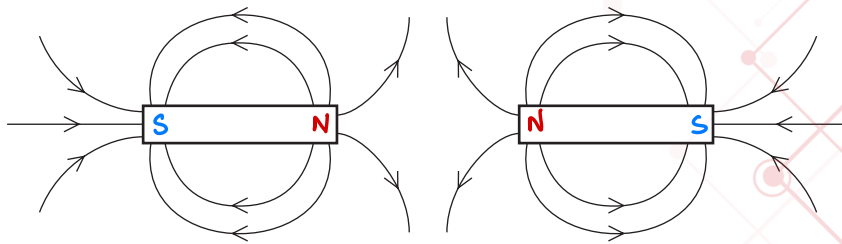
[MJ2013/P11/Q26]

- 04** The diagrams show a **small compass** close to a **strong bar magnet**. Which diagram shows the **correct compass direction**?

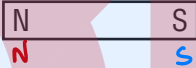


[MJ2013/P11/Q27]

05 The magnetic field around two bar magnets is shown.



Which diagram represents the correct arrangement of magnetic poles?

- A  
- B  
- C  
- D  

[MJ2013/P12/Q29]

06 A metal bar PQ hangs from a thin thread and always comes to rest with end P pointing north. Another bar XY of the same metal settles in no definite direction.

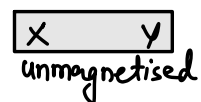
What happens if the two bars are brought near one another?

A End P and end Q both attract end X. ✓

B End P attracts end X but repels end Y. ✗

C End P neither attracts nor repels end X. ✗

D End P repels end X but attracts end Y. ✗



[ON2014/P11/Q26]

07 What is a suitable metal for the core of an electromagnet?

- A aluminium
- B copper
- C iron
- D steel

[ON2015/P11/Q23]

08 Which material can be picked up by a magnet?

A aluminium ✗

B copper ✗

C iron

D plastic ✗

[ON2015/P12/Q28]

Mark Scheme

ANSWER KEY

1	A
2	C
3	B
4	C
5	C
6	A
7	C
8	C

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01

- (a) Suggest a material that is used to make a permanent magnet.

Steel.

[1]

- (b) When there are no other magnetic fields present, the needle of a plotting compass points due north in the Earth's magnetic field. This is shown in Fig. 1.1.

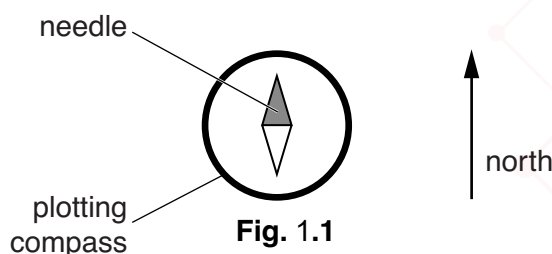


Fig. 1.1

The plotting compass is placed close to a permanent magnet, first at position A and then at position B, as shown in Fig. 1.2.

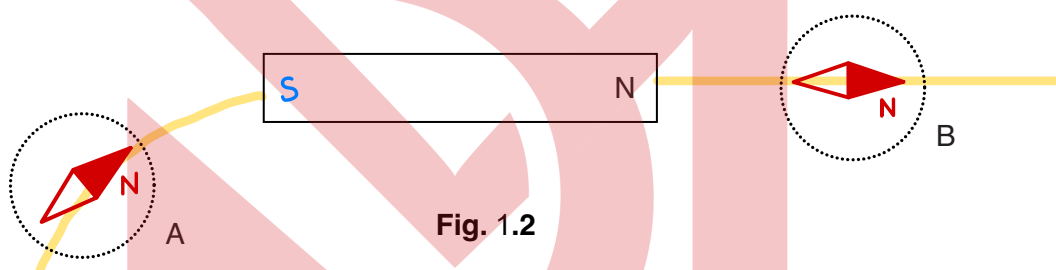


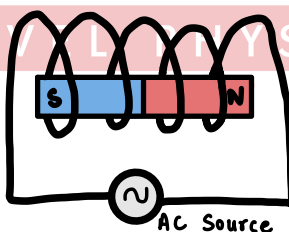
Fig. 1.2

The magnetic field close to the permanent magnet is very much larger than the magnetic field of the Earth.

On Fig. 1.2, draw needles in the two circles to show the direction in which the compass needle points when the compass is at A and at B.

[2]

- (c) Describe an electrical method that is used to demagnetise a permanent magnet. You may include a diagram in your answer.



Place magnet inside solenoid/coil & switch ON AC.

Gradually decrease the amount of voltage / Move the magnet out of solenoid slowly.

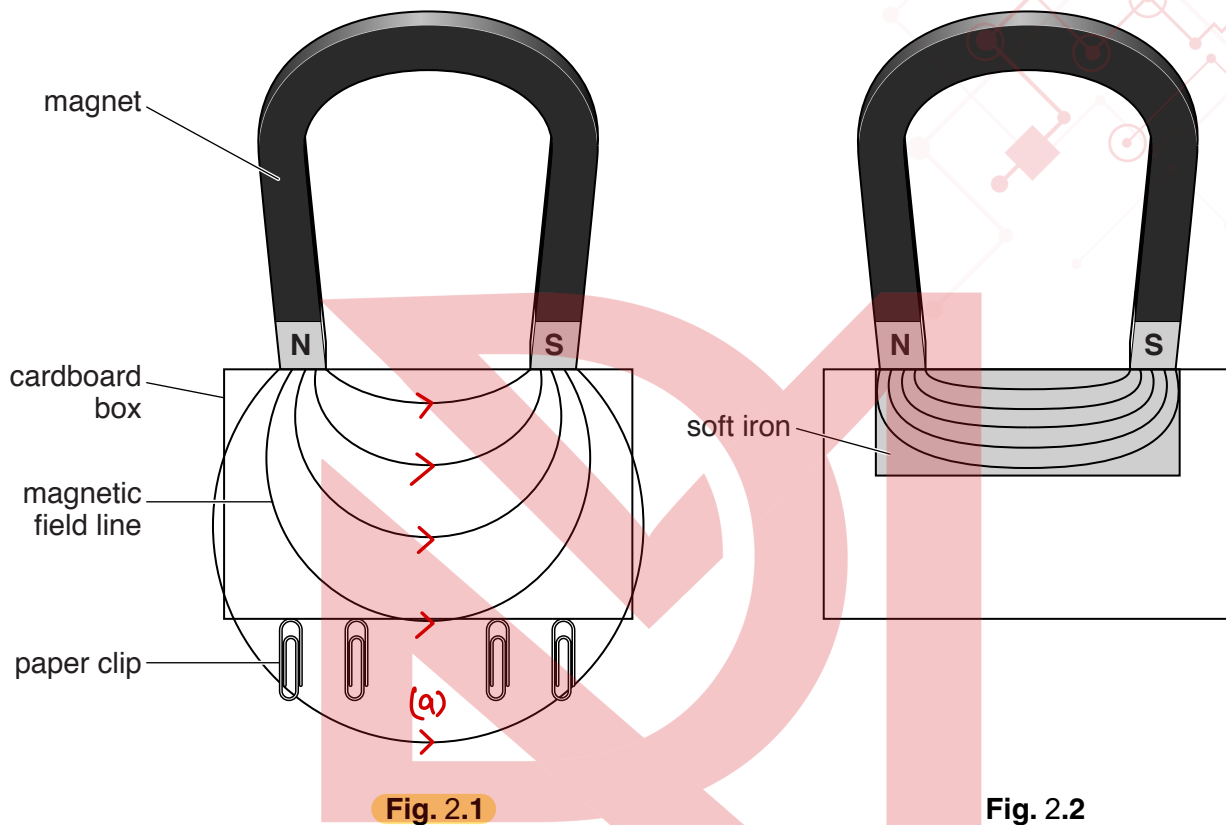
[3]

[ON2013/P21/Q6]

02 A teacher demonstrates magnetic screening. When a magnet is placed near a small cardboard box, paper clips on the other side of the box are picked up, as shown in Fig. 2.1.

When a small piece of soft iron is placed inside the box as shown in Fig. 2.2, the paper clips fall off.

Magnetic field lines in each diagram are shown as thin lines.



(a) On Fig. 2.1, mark an arrow on each of the magnetic field lines to show its direction. [1]

(b) Explain why placing the soft iron inside the box causes the paper clips to fall off.

All the field lines preferred to move through iron & none are left for paper clips so they fell off.

[2]

[MJ2014/P21/Q4]

- 03** Fig. 3.1 shows a compass needle near a bar magnet. Magnetic poles are shown on the compass needle and on the magnet.

A finger stops the compass needle from turning.

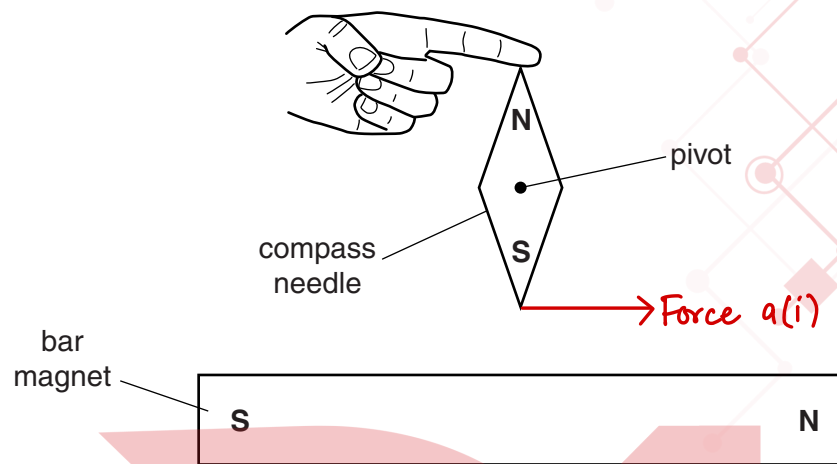


Fig. 3.1 (not to scale)

- (a) (i) The magnet causes a force on the S-pole of the compass needle.

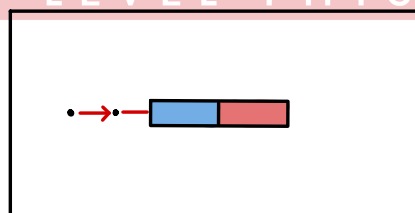
On Fig. 3.1, draw an arrow from the S-pole of the compass needle to show the direction of this force. [1]

- (ii) Explain why the compass needle turns when the finger is removed.

The force of attraction b/w south of compass & north of magnet rotated the needle about the pivot. [1]

- (b) A small compass is used to plot the magnetic field lines of the magnet.

Describe how the compass is used to plot magnetic field lines on a piece of paper.



Place bar magnet on paper & draw an outline around it. Place compass near the pole & mark a dot where compass needle aligns. Replace compass so that it aligns with dot & remark another dot. Join all the dots to draw magnet field lines.

[3]

[MJ2014/P22/Q7]

Mark Scheme

Page 3	Mark Scheme	Syllabus	Paper
	GCE O LEVEL – October/November 2013	5054	21

- 6 (a) steel/alnico/SmCo/NdFeB/magnetite B1
- (b) one needle fully correct **or** both angles correct – i.e. A bottom left to top right diagonal (0 < angle < 90°) **and** B horizontal C1
both needles fully correct (fully = angle and orientation) A1
- (c) (place) magnet in solenoid B1
a.c. supply to solenoid/coil (ignore cell/battery symbol) B1
withdraw magnet (slowly) **or** reduce current (slowly) B1 [6]

Page 3	Mark Scheme	Syllabus	Paper
	GCE O LEVEL – May/June 2014	5054	21

- 4 (a) arrow from N to S on at least three lines [B1]
- (b) magnetic field goes through soft iron **or** no field through clips [B1]
paper clips lose their (induced) magnetism [B1]
- [3]

Page 3	Mark Scheme	Syllabus	Paper
	GCE O LEVEL – May/June 2014	5054	22

- 7 (a) (i) **horizontal** arrow to right (by eye) [B1]
- (ii) forces/resultant causes moment **or** (turns because) force is not at pivot [B1]
- (b) mark made at one end/pole/direction of compass (on paper) [B1]
move compass so that other end of compass is on mark **and** remark [B1]
join marks made as compass moved on in some way (to draw line) [B1]
- [5]